

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2005

MATHEMATICS

Time:2 hours

READ THESE INSTRUCTIONS CAREFULLY

1. You have been given this question booklet and a separate sheet. The question booklet contains 50 question.
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only an ordinary pencil.
5. Make sure that have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number ?(i.e School Code Number and the figured Candidate's Number) in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answer are given. These answers are lettered A,B,C and D. In only ONE of the answers is correct .choose the correct answer .
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the question booklet:

9. what is the next number in the pattern:

1.2,2.4,4.8,____?

- A. 6.0
- B. 7.2
- C. 8.4
- D. 9.6

On the answer sheet:

The correct answer is D

9 [A] [B] [C] [D] 19 [A] [B] [C] [D] 29 [A] [B] [C] [D] 39 [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of the boxes

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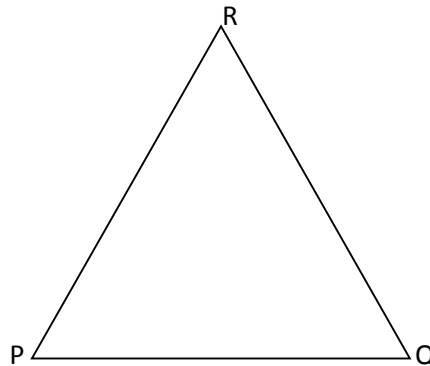
1. Which one of the following is 50205082 in words?
 - A. Fifty million two hundred and fifty thousand and eighty two
 - B. Fifty million twenty five thousand five hundred thousand and eighty two
 - C. Fifty million two hundred thousand five hundred and eighty two
 - D. Fifty million two hundred and five thousand and eighty two.
2. What is the place value of digit 6 in the number 2649053?
 - A. Hundred thousands
 - B. Millions
 - C. Six hundred thousand
 - D. Ten thousand
3. What is the number 29.34046 rounded off to the nearest thousandth?
 - A. 29.3
 - B. 29.34
 - C. 29.340
 - D. 29.3414
4. What is the smallest number that can be divided by 12,18 and 27 without a remainder?
 - A. 108
 - B. 36
 - C. 3
 - D. 5832
5. A shopkeeper bought 30 eggs. He sold each egg at sh.6, making a profit of 20 %. How much money had paid for the eggs?
 - A. Sh. 144
 - B. Sh. 150
 - C. Sh.180
 - D. Sh. 216
6. What is the value $8 \div 0.02 + 1.35 \times 0.4$?
 - A. 400.35
 - B. 40.54
 - C. 160.54
 - D. 400.54
7. The marked price of blouse was shs. 750. Halima bought five such blouses after being given a 10% discount. How much did she pay for the five blouses?
 - A. Sh. 3675
 - B. Sh. 675
 - C. Sh. 3 375
 - D. Sh. 3 700
8. Katua bought the following items from a kiosk
2kg of rice for shs 152
1 $\frac{1}{2}$ of meat @sh 160

2 loaves of bread @shs 23

What balances did he receive if he paid for the items using a shs. 1000 notes?

- A. Sh. 438
- B. Sh. 562
- C. Sh.410
- D. Sh.665

9. In the figure below, RPQ is a triangle. Point O is inside the triangle. Join RO,PO and QO.



What is the size of the obtuse angle POQ?

- A. 54°
- B. 114°
- C. 120°
- D. 126°

10. The age in the year of 10 pupils in a class are
16,18,15,14,17,16,14,13,19, and 14.

What is the median age of pupils?

- A. 16.5
- B. 15.6
- C. 15.6
- D. 15.5

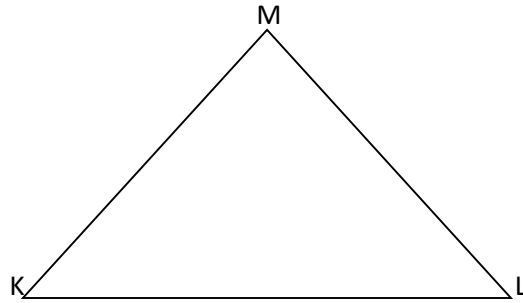
11. Which of the following statements is true about all quadrilaterals?

- A. Opposite side are equal
- B. Diagonals bisect at the right angles
- C. Sum of interior angles is 360°
- D. Angles are right angles

12. A container of volume 1 cubic metres is full of juice. The juice is then poured into five litres containers. How many such containers are used?

- A. 2
- B. 200
- C. 2000
- D. 20000

13. In the triangle KLM draw below, construct the perpendicular bisector of line KM to cut line KM at N and line KL at P?



What is the size of the acute angle KPN?

- A. 145°
 B. 90°
 C. 55°
 D. 35°
14. What is the value of x in $3(2x + 1) + 5(x + 4) = 61$?
- A. $3 \frac{5}{11}$
 B. $5 \frac{1}{11}$
 C. $7 \frac{7}{11}$
 D. $81 \frac{1}{11}$
15. Which one of the following is the correct net for an open cube

A.



E.

	B.	C.	D.
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16. The table below represent the scale of milk in litres by vendor in five days. The sale for Thursday is not shown.

Days	Mon	Tue	Wed	Thur	Fry
------	-----	-----	-----	------	-----

No of litres	25	19	23		22
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One litre of milk was sold for Sh. 25, the vendor got total of sh. 2925 for the sale of milk during the five days. How many litres of milk did the vendor sell on Thursday than on Tuesday?

- A. 9
 - B. 28
 - C. 47
 - D. 117
17. Sixteen workers can dig a field in 12 days. How many more workers are required so as to do the work in 8 days?
- A. 40
 - B. 24
 - C. 8
 - D. 10
18. The top of a ladder 26 metres long, leans on a vertical wall. The ladder touches the wall at a height of 10 metres. What is the horizontal distance from the bottom of the ladder to the wall?
- A. 16m
 - B. 18m
 - C. 24m
 - D. 576
19. The table below shows the postal rates for sending a money order

School fees for two children in the same school was sh. 9 400 and sh. 11 800. The father bought one money order to pay the total amount of fees. How much more would he have spent had he bought two separate money orders for the fees?

- A. Sh. 119
 - B. Sh.146
 - C. Sh.736
 - D. Sh. 617
20. What is the next number in the pattern?
1,3,7,15 ____?
- A. 2
 - B. 4
 - C. 8
 - D. 31
21. The table below shows Sagana-Nanyuki train fares

Value of order in sh	Commission
Not exceeding 500	Sh. 42
501 -1000	Sh. 114
1001 -3000	Sh. 174
3001 -5 000	Sh. 209
5001 -10 000	Sh. 295
10 001 -20 000	Sh. 441
20 001 -30 000	Sh. 617

The fare for the children is half that of adults. Taipei and his two children travelled from Nanyuki to Ssgana. On their return journey they first paid fare to Nanyuki. How much more money did they spend on travel for their return journey?

- A. Sh. 150
 B. Sh. 140
 C. Sh. 15
 D. Sh. 10
22. The figure below is a trapezium. Line $HG = 11\text{cm}$, $EF = 15\text{cm}$ and $HE = 3\text{cm}$. Line EF is parallel to HG and angle $FEH = \text{angle } EHG = 90^\circ$

Station	Nanyuki	Naromoro	Kiganjo	Karatina	sagana
Nanyuki					
Naromoro					
Kiganjo					
Karatina					
Sagana					

the area of the figure?

- A. 78cm^2
 B. 39cm^2
 C. 65cm^2

- D. 75cm^2
23. A rectangular container 45cm long and 25cm wide was full of water. After removing 22.5 litres of water, the level of water become 4cm high. What is the height of the container?
- A. 24cm
B. 20cm
C. 16cm
D. 6cm
24. Nekesa has p pencils. Rhoda has 3 more pencils than Nekesa. Karani has 2 pencils less than the total number that Nekesa and Rhoda have. How many pencils do they have altogether?
- A. $8p - 2$
B. $4p + 4$
C. $2p + 1$
D. $2p + 4$
25. Cherono spent Sh. 8 100 on food in May. In the month of June she spent 10% less on food than in May. How much money did she spent on food in June?
- A. Sh. 810
B. Sh. 7 290
C. Sh. 9 000
D. Sh. 8 910
26. Amina shared money among her three children. The first got $\frac{1}{3}$ while the second got $\frac{1}{4}$ of the money. The third got $\frac{1}{2}$ of what remained. Amina was left with sh. 1 500. How much money had she before its shared.
- A. Sh. 3000
B. Sh. 5700
C. Sh. 7200
D. 3 600
27. A motosist traveling at an evarege speed of 84 km/h took 2 hours and 30 minutes to travel from town M to town N. She then took 3 hours and 20 minutes to travel back to town M. What was the average speed for the whole journey?
- A. 36km/h
B. 63km/h
C. $73 \frac{1}{2}$ km/h
D. 72km/h
28. In the figure below, Line PQ is parallel to line SR and QT is equal to TR. Angle QPT = 31° and angle TRS = 57°

What is the size of angle PQR?

- A. 46°
B. 88°

C. 103^0

D. 92^0

29. The charges of sending a telegram were as follows

The first 10 words or less sh. 15

Any additional words sh. 1.50 each

Abbreviations and punctuation mark are counted as words

A tax of 20% is charged on the amount to be paid is rounded off to the nearest 50 cents. What was the cost of sending the following telegram?

JOHN MLAMA P.O. BOX 360 NYERI GOING TO KISUMU AFTER THE EXAMINATION KOIGI

A. Sh. 25

B. Sh. 24

C. Sh. 28.80

D. Sh. 29.00

30. Below is travel graph showing the journey of the motorist traveling from town L to town K and back, and that of the cyclist traveling from town L to town K.

How far from town L was the cyclist when he met the motorist traveling back to town L?

A. 40km

B. 32km

C. 20km

D. 8km

31. What is the value of $\frac{1}{3}(2x + 4y^2) + 5p - 8$

When $p=6$, $x=2p$, and $y = \frac{1}{2}x - 3$?

A. 130

B. $63 \frac{1}{3}$

C. $43 \frac{1}{3}$

D. $36 \frac{2}{3}$

32. A machine packs 250 two kilogram packets of sugar while another packs 375 tones of sugar altogether, do the two machines pack in 5 days?

A. 0.875

B. 3.125

C. 4.375

D. 6.25

33. A square of side 2cm is cut from each corner of rectangular paper measuring 8 cm by 12cm.

Which one of the following statements is true?

A. The perimeter of the following paper is greater than the perimeter of the rectangle paper by 16cm

B. The perimeter of the remaining paper is less than the perimeter of the rectangle paper by 16cm

C. The perimeter of the rectangular paper is reduced by 8cm

D. The perimeter of the remaining paper is equal to the perimeter of rectangle paper.

34. A salesman is paid a salary of sh. 1 500 per month plus commission. On the month of June he get a commission of $2\frac{1}{2}\%$ on the sale of goods above sh. 10,000. In one month he was paid a total of sh. 1800. How much was the sale of the goods?
- Sh. 300
 - Sh. 12,000
 - Sh. 22,000
 - Sh. 82,000
35. In a group of 126 spectators the ratio of men to women was 3:4. What is the new ratio joined the group of spectators?
- 7:10
 - 5:12
 - 13:16
 - 1:4
36. A clock was set on Monday at 8.30 a.m. on Tuesday the following day the clock showed 8.45p.m. when the correct time was 8.30p.m. How many minutes was the clock gaining in every 24 hours?
- 10 minutes
 - $7\frac{1}{2}$ minutes
 - 15 minutes
 - 30 minutes
37. The base of the closed cuboid measures 4cm by 5cm and the height is 7cm. The base and the top parts of a cuboid are painted. What is the total surface area of the parts which are not painted?
- 166cm^2
 - 126cm^2
 - 63cm^2
 - 40cm^2
38. During an election the winning candidates got 0.425 of the votes cast while the other two candidates got 0.39 and 0.183 respectively. There were 48 spoilt votes. How many votes did the winning candidate get?
- 24,000
 - 10 200
 - 9 360
 - 4 392
39. A motorist started a journey of 250km at 6.30a.m. traveling at an average speed of 100km/h. After travelling for 150 km, the car got a punctured and it took him 30 minutes to change the wheel. He then continued with the rest of the journey at an average speed of 80km/h. At what time did he reach his destination?
- 9.15a.m
 - 9.45a.m
 9. 30a.m
 - 9.55a.m

40. Four children bought 53 oranges altogether. Nekoye bought x oranges and Kamau bought 9 oranges more than Nekoye. Fatuma bought twice as many oranges as Nekoye. Otieno bought as many oranges as both Kamau and Fatuma bought. Which one of the following equations can be used to find the number of oranges Nekoye bought?
- A. $5x + 22 = 53$
 - B. $5x + 18 = 53$
 - C. $7x + 9 = 53$
 - D. $7x + 18 = 53$
41. Waithera borrowed sh. 10 000 for the period of two years. She was charged compound interest at the rate of 15% per year. How much interest did she pay altogether?
- A. Sh. 1 500
 - B. Sh. 3 000
 - C. Sh. 3 225
 - D. Sh. 1 725
42. The figure below represents a table mart made up of a rectangle and four semi-circles. The triangle measures 14cm by 7cm.

What is the area of the table- mart?

(Taken 22/7)

- A. $192 \frac{1}{2} \text{ cm}^2$
 - B. 862 cm^2
 - C. 290 cm^2
 - D. 194 cm^2
43. The cash price of tractor is 1.8 million shillings Rotich bought on hire purchase term. The total amount he paid was 30% more than the cash price. He deposit of sh. 600 000 and the remainder in 24 equal instalments. How much was each instalments?
- A. Sh. 70 000
 - B. Sh. 97,000
 - C. Sh. 47 000
 - D. Sh. 25,000
44. Omala and Mwite had packets of tea to be packed into cartons. Each carton holds 46 packets. Omala packed 63 cartons and remained with 24 packets while mwite packed 54 cartons and remained with 19 packets. How many more packets of tea had Omala than Mwite?
- A. 419
 - B. 414
 - C. 409
 - D. 5425
45. The figure shown below is formed by two right triangles ABD and BCD

What is the area of the triangle BCD?

- A. 300 cm^2
- B. 180 cm^2

- C. 84cm^2
- D. 150cm^2

46. The population of the village is represented by the pie chart below.

If there were 300 girls, how many more boys than men were there?

- A. 80
- B. 100
- C. 180
- D. 50

47. Which is the correct order of writing the fraction $\frac{3}{4}$, $\frac{7}{9}$, $\frac{4}{5}$ and $\frac{9}{11}$ from the largest to the smallest?

- A. $\frac{3}{4}$, $\frac{4}{4}$, $\frac{7}{9}$, $\frac{9}{11}$
- B. $\frac{9}{11}$, $\frac{7}{9}$, $\frac{4}{5}$, $\frac{3}{4}$
- C. $\frac{9}{11}$, $\frac{7}{9}$, $\frac{4}{5}$, $\frac{4}{5}$
- D. $\frac{9}{11}$, $\frac{4}{5}$, $\frac{7}{9}$, $\frac{3}{4}$

50.

Which one of the shape below would fit in the dotted space in the pattern above?